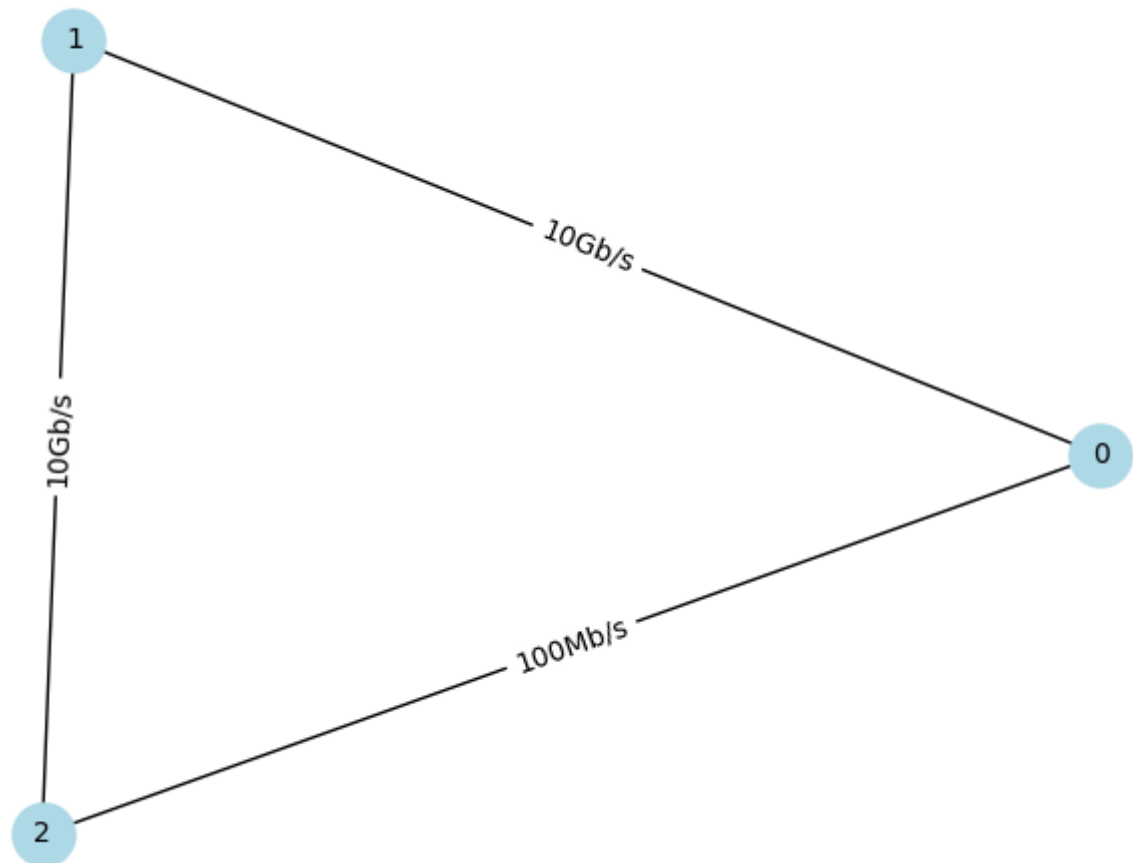


Spanning Tree Protocol

Seed: 1999025788



Consider the network in the figure and assume that:

- The adopted algorithm is the **Spanning Tree Protocol (STP)**.
- BPDUs are formatted as a tuple: (Root ID, Path Cost, Sender ID).
- Tie-breaking preference is: **Lowest Root ID, Lowest Cost, Lowest Sender ID**.

Instructions:

Write down the sequence of message exchanges. For every BPDU that causes a state change in the receiver, write:

1. The Sender -> Receiver and the [BPDU] content.
2. The receiver's **Updated Best BPDU**.

3. The receiver's **Updated Port List** using the format
[Neighbor:ROLE].

Use the following abbreviations for roles: ROOT, DESI (Designated),
and BLOC (Blocked).

Message Sequence

0 -> 1: [BPDU] (0, 0, 0)

Updated BPDU of 1: (0, 2000, 1)

Updated Ports of 1: [0:ROOT, 2:BLOC]

0 -> 2: [BPDU] (0, 0, 0)

Updated BPDU of 2: (0, 200000, 2)

Updated Ports of 2: [0:ROOT, 1:BLOC]

1 -> 2: [BPDU] (0, 2000, 1)

Updated BPDU of 2: (0, 4000, 2)

Updated Ports of 2: [0:BLOC, 1:ROOT]

Final Network State

Node 0

Cost: 0

Port State: [1:DESI, 2:DESI]

Node 1

Cost: 2000

Port State: [0:ROOT, 2:DESI]

Node 2

Cost: 4000

Port State: [0:BLOC, 1:ROOT]